Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/27/2012
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2004	**End Date:**	12/01/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2012: David Ware was removed from the protocol. October 2010: Buccello-Stout, Regina (Universities Space Research Association) Removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Yarbough, Patrice (Universities Space Research Association) Zanello, Susana (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Flight Analogs Project Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume, and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In FY2012, standard measures were collected in two bed rest campaigns: Campaign 11 (C11) and Campaign 17 (C17). C11 is an ongoing campaign that examines physical function and countermeasure effects that result from 70-days of 6-degree head-down tilt bed rest. Four investigations are integrated in this study complement. One study identifies the key underlying physiological factors that contribute to changes in performance using a set of functional task tests that are representative of critical mission tasks for planetary operations. A second study examines the ability of an integrated resistive and aerobic training (iRAT) program to optimally mitigate the deleterious effects of long duration head-down tilt bed rest. A third study adds the use of a physiological dose of testosterone with the iRAT exercise program to enhance the effectiveness of exercise in preserving muscle and bone structure during long duration bed rest. The fourth study examines the relationship of smelling acuity and nasal congestion that result from six degree head-down tilt bed rest and how this impacts food acceptability and menu boredom. Results of these studies will help to optimize exercise countermeasures (with and without supplementation) and determine their effects on both physical and physiologic functioning after long duration bed rest. C17 utilized the 6-degree head-down tilt bed rest model to simulate cardiovascular changes that occur during spaceflight. Some of these changes include reductions in plasma volume that can lead to orthostatic intolerance (lightheadedness and fainting). This short duration bed rest study tested whether a three piece, custom fit compression garment prevented orthostatic intolerance after 14 days of 6° head-down tilt bed rest. In addition, a schedule for progressive removal of this garment was studied to facilitate re-adaptation to the vertical position after bed rest. A total of 16 subjects were studied and divided into two groups. The first group (control group) wore the chest-high compression garment during testing on the morning of the day that they got up from bed (BR+0). The garment was removed after testing and the subject completed the re-adaptation process without garments. A second group (experimental group) wore the chest-high garment for the entire first day out of bed and wore thigh high garments for the next 3 days thus, gradually reducing compression over the course of 3 days. Results of this study indicated that wearing compression garments on BR+0 prevented orthostatic intolerance. Use of the garments on subsequent days should be done when needed to assist re-adaptation. Standard measures results from this campaign are highlighted below. Plasma Volume Plasma volume provides an indication of the fluid loss due to the 6-degree head-down tilt position. Results in this study are similar to other bed rest studies. Plasma volume decreased from pre-bed rest (BR-5) to post bed rest (BR+0), but appears to recover by the fourth day after bed rest (BR+3). Compression garments worn by the countermeasure subjects for three days post-bed rest do not appear to prevent recovery of plasma volume. Cardiovascular Response to Tilt The tilt test is used to assess orthostatic tolerance before and after bed rest. There was no significant difference in blood pressure for either the control or experimental (countermeasure) subjects on the day they got up from bed (BR+0) as compared to pre-bed rest (BR-5) values. This suggests a protective effect of the compression garments against orthostatic intolerance. In previous studies, subjects who did not wear compression garments demonstrated an increase in heart rate. An attenuated heart rate response was observed in this study by subjects who wore compression garments on BR+0. Heart rate values on BR+0 compared well to BR-5, suggesting that the compression garments were protecting against orthostatic tachycardia during upright tilt. Differences were not detected between the countermeasure and control groups on BR+3. Aerobic Capacity and Muscle Strength Cycle ergometry is used to implement a graded exercise test where subjects exercise to maximal effort. Maximum oxygen uptake (VO2peak) provides an assessment of aerobic capacity for bed rest subjects. While there is a decreasing trend in VO2peak, there was no significant difference between control and experimental groups from pre- to post bed rest. The short duration of the bed rest phase (14-days) was not long enough to affect a significant change in these subjects. Similar trends were seen for muscle strength assessed by isokinetic muscle testing. Knee extensor strength

	(KES) and knee flexor strength (KFS) did not demonstrate a significant difference from pre- to post bed rest. T-Reflex and Postural Control T-Reflex is a stretch reflex is assessed in the calf muscles of the leg. Monitoring this reflex provides information on changes in peripheral neural control of muscles that control upright posture. This test can be completed on subjects during the bed rest phase. No significant differences were found for parameters of stretch reflex when compared to the pre-bed rest baseline values. In addition, there was no difference between subject groups at each measurement time point. Assessment of postural control demonstrated decrements in performance on BR+0. This was particularly true for the most challenging task where vision was removed and proprioception was challenged while subjects moved their heads. These postural control decrements quickly resolved as subject returned to baseline performance by BR+4. There was no difference between control and experimental groups.
Bibliography Type:	Description: (Last Updated: 05/16/2019)
Articles in Peer-reviewed Journals	Spatz JM, Fields EE, Yu EW, Pajevic PD, Bouxsein ML, Sibonga JD, Zwart SR, Smith SM. "Serum sclerostin increases in healthy adult men during bed rest." J Clin Endocrinol Metab. 2012 Sep;97(9):E1736-40. Epub 2012 Jul 5. PMID: 22767636 , Sep-2012
Articles in Peer-reviewed Journals	Smith SM, Heer M, Wang Z, Huntoon CL, Zwart SR. "Long-duration space flight and bed rest effects on testosterone and other steroids." J Clin Endocrinol Metab. 2012 Jan;97(1):270-8. Epub 2011 Nov 2. PMID: 22049169 , Jan-2012